Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-4536
Project Name	LegalEase – AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the LegalEase system behaves under expected user load, especially during AI-powered document generation, document preview, editing, and file download operations. The main backend API is the FastAPI /generate endpoint, which accepts document type, parties, terms, and dates and returns generated legal content through Gemini.

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Postman / Manual API Testing
Type of Testing	Load Testing / Response Time Testing
Target Module / API	FastAPI /generate API and Legal Document Generation
Test Environment	Local Development Environment
Test Date	15 March 2026

The /generate API processes user inputs such as document type, parties, terms, and effective dates and sends them to the Gemini document generator.

Step 2: Test Scenarios

S.No	**Test Scenario / Description**	**No. of Virtual Users**	**Duration (sec)**	**Expected Outcome**
1	Generate an Employment Contract using valid user inputs	1	60	Legal document should be generated successfully
2	Generate an NDA using	5	60	API should process

	parties, terms and effective date			requests and return generated content
3	Generate a Lease Agreement with multiple terms	10	60	System should handle multiple requests without critical errors
4	Generate and download documents in TXT, DOCX and PDF formats	10	60	Documents should be generated and downloaded correctly

LegalEase supports document generation for agreements, contracts, NDAs, lease agreements and employment documents, with TXT, DOCX and PDF export options.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	Pending	Measure average /generate response time
2	Response Time (Max)	< 5 seconds	To be measured	Pending	Measure maximum response time
3	Throughput (Req/sec)	To be measured	To be measured	Pending	Measure requests processed per second

4	Error Rate	< 1%	To be measured	Pending	Check failed API requests
5	CPU Utilization	< 80%	To be measured	Pending	Monitor CP